

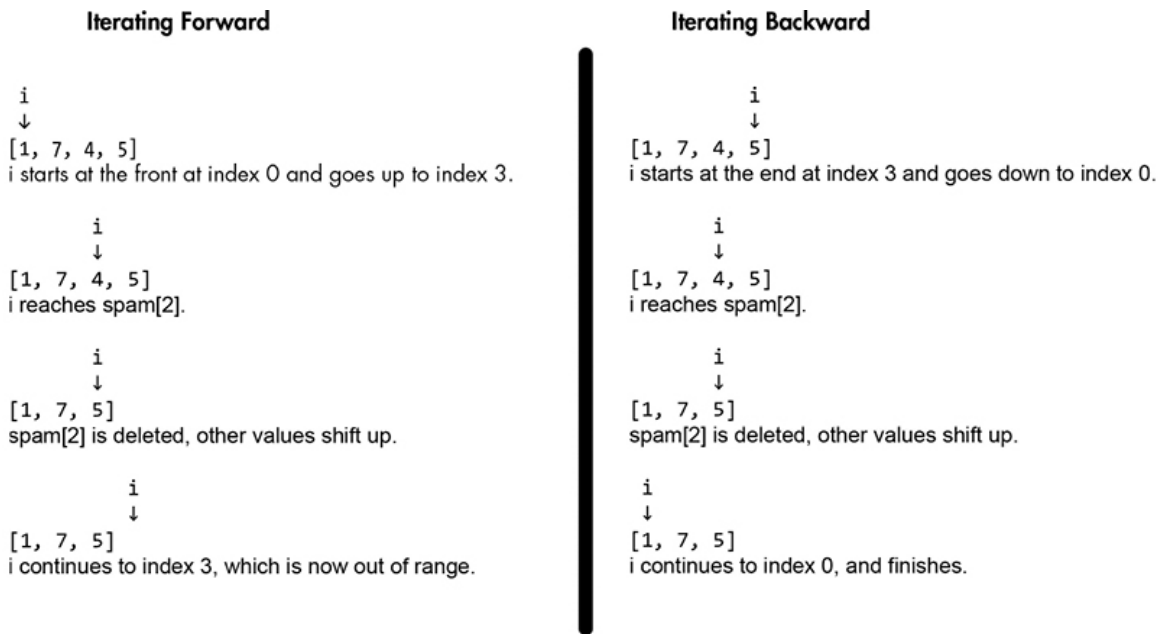


Figure 8-3: Removing even numbers from a list while iterating forward (left) and backward (right)

Similarly, you can add items to the end of the list as you iterate backward over it. Enter the following into the interactive shell, which appends a copy of any even integers in the `someInts` list to the end of the list:

```
>>> someInts = [1, 7, 4, 5]
>>> for i in range(len(someInts) - 1, -1, -1):
...     if someInts[i] % 2 == 0:
...         someInts.append(someInts[i])
...
>>> someInts
[1, 7, 4, 5, 4]
```

A visualization of the execution of this code is at <https://autbor.com/iteratebackwards2>. By iterating backward, we can append items to or remove items from the list. But this can be tricky to do correctly because slight changes to this basic technique could end up introducing bugs. It's much simpler to create a new list rather than modifying the original list. As Python core developer Raymond Hettinger put it:

Q. What are the best practices for modifying a list while looping over it?